Junseok Ahn

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Daejeon - 34184, South Korea

Research Interests

My research focuses on multimodal learning and generative modeling for natural human–AI interaction, particularly joint audio–visual generation and efficient streaming conversational agents. I am also interested in evaluating the quality, cross-modal consistency, and contextual appropriateness of generated speech and facial behavior.

Education

- **Korea Advanced Institute of Science and Technology (KAIST)** *Mar 2023 – Present*
  Integrated MS/PhD in Electrical Engineering Daejeon, South Korea
  - Advisor: Joon Son Chung
  - Expected graduation: Feb 2028
- **Korea Advanced Institute of Science and Technology (KAIST)** *Mar 2019 – Feb 2023*
  BS in Electrical Engineering Daejeon, South Korea
  - Summa Cum Laude; ranked 2nd among KAIST's 2023 bachelor's graduates
  - GPA: 4.17/4.3

Publications

C=Conference, J=Journal, S=Submitted

[S.4] **DIAL-S2ST: Learning Latency Control for Simultaneous Speech-to-Speech Translation**
Jeongsoo Choi, **Junseok Ahn**, Joon Son Chung
arXiv preprint, 2026

[S.3] **X-Dubbing: Speech-to-Speech Translation under Temporal and Visual Constraints**
Jeongsoo Choi, Ji-Hoon Kim, **Junseok Ahn**, Joon Son Chung
arXiv preprint, 2026

[S.2] **JAVIS: Joint Audio-Visual Interaction and Streaming for Real-Time Conversational Agents**
Junseok Ahn, Jaehyun Park, Jeongsoo Choi, Joon Son Chung, Ji-Hoon Kim
arXiv preprint, 2026

[S.1] **TAVID: Text-Driven Audio-Visual Interactive Dialogue Generation**
Ji-Hoon Kim*, **Junseok Ahn***, Doyeop Kwak, Joon Son Chung, Shinji Watanabe
arXiv preprint, 2025

[J.1] **Deep Understanding of Sign Language for Sign to Subtitle Alignment**
Youngjoon Jang, Jeongsoo Choi, **Junseok Ahn**, Joon Son Chung
IEEE Transactions on Multimedia, 2026

[C.4] **VoiceDiT: Dual-Condition Diffusion Transformer for Environment-Aware Speech Synthesis**
Jaemin Jung*, **Junseok Ahn***, Chaeyoung Jung, Tan Dat Nguyen, Youngjoon Jang, Joon Son Chung
ICASSP, 2025 · Oral presentation

[C.3] **VoxSim: A perceptual voice similarity dataset**
Junseok Ahn, Youkyum Kim, Yeunju Choi, Doyeop Kwak, Ji-Hoon Kim, Seongkyu Mun, Joon Son Chung
Interspeech, 2024 · **Best Student Paper Award Nominee**

[C.2] **Faces that Speak: Jointly Synthesising Talking Face and Speech from Text**
Youngjoon Jang*, Ji-Hoon Kim*, **Junseok Ahn**, Doyeop Kwak, Hong-Sun Yang, Yoon-Cheol Ju, Il-Hwan Kim, Byeong-Yeol Kim, Joon Son Chung
CVPR, 2024

[C.1] **Slowfast Network for Continuous Sign Language Recognition**
Junseok Ahn*, Youngjoon Jang*, Joon Son Chung
ICASSP, 2024

* denotes equal contribution.

Experience

- **IITP-funded Project: Sign Language Video Anonymization** *Apr 2025 – Present*
  Graduate Researcher, KAIST Multimodal AI Lab Daejeon, South Korea
  - Develop avatar-based video anonymization methods for sign language communication, with a focus on preserving hand motion, body pose, and facial expressions.
  - Design 3D motion parameter prediction models and integrate them with 3D Gaussian Splatting-based avatar rendering, targeting real-time motion transfer.

- **Samsung Research–KAIST Collaboration [🌐]** *Jul 2023 – Dec 2023*
  Graduate Researcher, KAIST Multimodal AI Lab Daejeon, South Korea
  - Led data curation, perceptual annotation, and quality control for VoxSim (*Interspeech*, 2024).
  - Developed and evaluated WavLM- and ECAPA-TDNN-based models for perceptual speaker similarity and synthesized speech quality assessment.

- **KAIST Multimodal AI Lab [🌐]** *Mar 2022 – Dec 2022*
  Undergraduate Researcher, Undergraduate Research Program (URP) Daejeon, South Korea
  - Investigated language-invariant speaker embeddings using adversarial training with gradient reversal, and evaluated cross-lingual speaker verification.

- **Samsung Electronics, System LSI [🌐]** *Sep 2021 – Feb 2022*
  Student Researcher (Intern) South Korea
  - Developed a MATLAB simulator for a 5G NR PUSCH MIMO receiver.
  - Implemented and compared zero-forcing, LMMSE, and maximum-likelihood receiver algorithms through bit error rate evaluation.

Academic Service

- **Conference Reviewer:** SLT 2026, EMNLP 2026, ICML 2026, ICASSP 2026, ASRU 2025, NeurIPS 2025

Skills

- **Programming:** Python, MATLAB
- **Frameworks and Libraries:** PyTorch, Hugging Face (Transformers, Diffusers, Accelerate)
- **Audio and Video Processing:** FFmpeg, OpenCV, torchaudio, librosa, Decord, MediaPipe

Honors and Awards

- **Best Teaching Assistant Award** *Mar 2025*
  KAIST School of Electrical Engineering
  - EE488 Deep Learning for Visual Understanding (Fall 2024).

- **Dean's List** *Sep 2021, Sep 2020, Sep 2019*
  KAIST
  - Recognized for academic excellence in Spring 2021, Spring 2020, and Spring 2019.

Teaching Experience

- **EE738 Speech Recognition Systems** *Spring 2026, Spring 2024, Spring 2023*
  Teaching Assistant, KAIST
  - Served as head teaching assistant in Spring 2024.

- **EE488 Deep Learning for Visual Understanding** *Fall 2024, Fall 2023*
  Teaching Assistant, KAIST
  - Served as head teaching assistant in Fall 2024.

Additional Information

Languages: Korean (native), English (fluent)

References

1. **Joon Son Chung**
   Associate Professor, School of Electrical Engineering
   Korea Advanced Institute of Science and Technology (KAIST)
   Email: joonsonchung@gmail.com
   Relationship: PhD advisor